Qualification Pack

Infrastructure Technician - 5G Networks

QP Code: TEL/Q4201

Version: 3.0

NSQF Level: 4

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Contents

Qualification Pack

TEL/Q4201: Infrastructure Technician - 5G Networks

Brief Job Description

An Infrastructure Technician - 5G Networks is responsible for installing the passive infrastructure equipment at 5G network infrastructure sites to ensure power supply to 5G network and transmission equipment. The individual is also responsible for maintaining the passive infrastructure equipment. The person may also carry out regular repair and maintenance of the 5G network infrastructure.

Personal Attributes

The individual should be physically fit to work for long durations. The person should have an aptitude for details with problem-solving skills and the ability to work in coordination with others. The individual should be able to communicate appropriately, both verbally and in writing.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. TEL/N4202: Prepare for the Installation of 5G Network Infrastructure
2. TEL/N4203: Install and Set up Passive Infrastructure Equipment with the 5G Equipment
3. TEL/N4204: Maintain Passive Infrastructure Equipment
4. TEL/N9109: Follow sustainable practices in telecom infrastructure management
5. DGT/VSQ/N0101: Employability Skills (30 Hours)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Network (Passive) Installation
Country	India
NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3114.1301

Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Electronics Engineering/ Electrical Engineering/ Wireless Communication Technology/ Networking and Mobile Communications) OR 10th grade pass with 3 Years of experience in the operations and maintenance of passive infrastructure OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience in the operations and maintenance of passive infrastructure OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in the operations and maintenance of passive infrastructure
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-TL-04084-2025-V2-TSSC
NQR Version	2

TEL/N4202: Prepare for the Installation of 5G Network Infrastructure

Description

This unit is about making appropriate arrangements for installing the relevant 5G network infrastructure at the identified sites

Scope

The scope covers the following :

- Conduct the site survey
- Prepare for the installation of 5G network infrastructure

Elements and Performance Criteria

Conduct the site survey

To be competent, the user/individual on the job must be able to:

PC1. determine the scope of work, including the identification of the site selected for the installation of 5G network infrastructure

PC2. coordinate with relevant personnel to conduct a site survey and assess the site's suitability for installation

PC3. inspect the site for accessibility, structural integrity, and potential safety hazards such as electrical and mechanical risks

PC4. identify and document site-specific modifications required for installation and coordinate with relevant personnel for necessary changes

PC5. assess environmental conditions that may impact infrastructure performance and recommend mitigation measures

PC6. verify the availability of power supply and backhaul connectivity at the site for seamless network integration

Prepare for the installation of 5G network infrastructure

To be competent, the user/individual on the job must be able to:

PC7. obtain the required authorizations and complete the necessary paperwork before installation

PC8. arrange and verify the availability of equipment, accessories, and tools required for installation

PC9. inspect all equipment, accessories, and tools for functionality, and replace faulty or damaged items

PC10. assemble and configure networking gear and infrastructure components as per installation requirements

PC11. coordinate with relevant personnel for the erection of 5G cell towers and positioning of related equipment

PC12. document all pre-installation activities, including site readiness, equipment inventory, and compliance requirements

PC13. ensure safety measures are in place and communicate installation plans with relevant stakeholders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant documents required for network infrastructure preparation activities
KU2. equipment, accessories, tools, and safety gear required for 5G network infrastructure installation
KU3. importance of pre-installation checks to ensure all equipment is functional and meets industry standards
KU4. process of assembling and configuring network gear for seamless infrastructure integration
KU5. authorization and regulatory compliance requirements for telecom infrastructure installation
KU6. installation techniques, including positioning, securing, and grounding of telecom equipment
KU7. environmental and structural considerations affecting 5G network performance and reliability
KU8. site-specific risk assessment and mitigation strategies to ensure safe installation
KU9. documentation standards and record-keeping practices for telecom infrastructure installation activities
KU10. workplace safety protocols and best practices for handling telecom equipment and working at heights

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain accurate and detailed documentation of work-related activities
GS2. read and interpret technical documents, installation guidelines, and regulatory standards
GS3. listen attentively and comprehend work instructions for accurate execution
GS4. communicate professionally with team members, supervisors, and stakeholders
GS5. prioritize tasks effectively to ensure timely and efficient completion of work
GS6. collaborate with co-workers and stakeholders to facilitate smooth installation processes
GS7. identify challenges and evaluate potential solutions for effective problem-solving
GS8. make prompt decisions to address workplace emergencies and mitigate risks
GS9. adapt to new tools, technologies, and industry trends relevant to 5G infrastructure installation
GS10. follow workplace safety protocols to prevent accidents and ensure compliance with industry regulations

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Conduct the site survey	**15**	**25**	**-**	**10**
PC1. determine the scope of work, including the identification of the site selected for the installation of 5G network infrastructure	2.5	4	-	1.5
PC2. coordinate with relevant personnel to conduct a site survey and assess the site's suitability for installation	2.5	4	-	1.5
PC3. inspect the site for accessibility, structural integrity, and potential safety hazards such as electrical and mechanical risks	2.5	4.5	-	2
PC4. identify and document site-specific modifications required for installation and coordinate with relevant personnel for necessary changes	2.5	4.5	-	1.5
PC5. assess environmental conditions that may impact infrastructure performance and recommend mitigation measures	2.5	4	-	2
PC6. verify the availability of power supply and backhaul connectivity at the site for seamless network integration	2.5	4	-	1.5
Prepare for the installation of 5G network infrastructure	**15**	**25**	**-**	**10**
PC7. obtain the required authorizations and complete the necessary paperwork before installation	2	3	-	1.5
PC8. arrange and verify the availability of equipment, accessories, and tools required for installation	2	4	-	1.5
PC9. inspect all equipment, accessories, and tools for functionality, and replace faulty or damaged items	2	4	-	1.5
PC10. assemble and configure networking gear and infrastructure components as per installation requirements	2.5	4	-	1.5

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. coordinate with relevant personnel for the erection of 5G cell towers and positioning of related equipment	2.5	4	-	1.5
PC12. document all pre-installation activities, including site readiness, equipment inventory, and compliance requirements	2	3	-	1.5
PC13. ensure safety measures are in place and communicate installation plans with relevant stakeholders	2	3	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4202
NOS Name	Prepare for the Installation of 5G Network Infrastructure
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Network (Passive) Installation
NSQF Level	4
Credits	5
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

TEL/N4203: Install and Set up Passive Infrastructure Equipment with the 5G Equipment

Description

This unit covers installing passive infrastructure equipment and integrating them with 5G equipment to ensure proper functionality and power management at telecom infrastructure sites.

Scope

The scope covers the following :

- Install passive infrastructure equipment
- Set up passive infrastructure with 5G equipment

Elements and Performance Criteria

Install passive infrastructure equipment

To be competent, the user/individual on the job must be able to:

PC1. select appropriate passive infrastructure equipment, such as battery bank, Switched-Mode Power Supply (SMPS) unit, Power Interface Unit (PIU), Diesel Generator (DG) set, and Climate Control Unit, based on site requirements

PC2. conduct tests on passive infrastructure equipment before installation to verify their functionality

PC3. install the battery bank, battery charger, and battery stand as per manufacturer's guidelines, ensuring sufficient batteries as per power requirements

PC4. charge and test the battery bank to confirm its ability to charge and discharge properly

PC5. install the SMPS unit according to manufacturer instructions and verify it provides regulated output voltage

PC6. install and test the PIU to ensure protection against voltage fluctuations and surges

PC7. identify a suitable location for the DG set, ensuring proper ventilation, obstruction-free airflow, and minimal dust accumulation

PC8. ensure adequate space for DG set operation, servicing, and ventilation

PC9. install the DG set following manufacturer instructions and implement air cleaners for dusty environments and anti-condensation heaters for humid conditions

PC10. install Climate Control Units to maintain optimal temperature and humidity levels for telecom equipment

PC11. ensure correct placement of all passive infrastructure equipment, adhering to manufacturer specifications

PC12. perform structured cabling for installed equipment, ensuring compatibility and adherence to power distribution guidelines

PC13. establish a power connection between telecom equipment and the power source to ensure safe and continuous operation

PC14. conduct functionality tests on installed infrastructure equipment and rectify any identified issues

Set up passive infrastructure with 5G equipment

To be competent, the user/individual on the job must be able to:

PC15. install the power supply unit for gNodeB/eNodeB, ensuring compatibility with the equipment

PC16. install a heat sink for gNodeB/eNodeB to facilitate heat dissipation and prevent overheating

PC17. connect the gNodeB/eNodeB with the power supply unit and heat sink using appropriate cabling, ensuring secure connections

PC18. establish a secure connection between power supply units and the main power source

PC19. set up a reliable power connection between the 5G base station and the power source, adhering to equipment power requirements

PC20. use appropriate cables for power supply and provide protection against environmental exposure and wear

PC21. connect and power up 5G base stations, 5G Radio Frequency (RF) transceivers, and Remote Radio Units (RRUs) following specified power guidelines

PC22. integrate emergency power supply systems with 5G network and transmission equipment

PC23. install protective shelters and telecom ducts to safeguard passive infrastructure equipment and cabling

PC24. establish a site-wide power distribution system for consistent power delivery to all cell tower equipment

PC25. use suitable Personal Protective Equipment (PPE) while handling electrical installations to ensure safety

PC26. conduct final testing of all equipment post-installation and troubleshoot any operational issues

PC27. coordinate with equipment manufacturers to address any defects or malfunctions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of passive infrastructure equipment used at telecom sites, including their functions and installation requirements

KU2. the significance of pre-installation testing for passive infrastructure equipment to confirm proper functionality

KU3. the procedures for installing and maintaining battery banks, chargers, and stands, ensuring adequate power backup

KU4. the requirements for testing battery performance and addressing charging issues

KU5. the installation procedures for SMPS units and tests to ensure regulated output voltage

KU6. the role and installation of PIU in protecting telecom equipment from power fluctuations

KU7. the importance of uninterrupted power supply for 5G network and transmission equipment

KU8. the factors for selecting installation locations for passive infrastructure equipment as per manufacturer guidelines

KU9. the considerations for DG set placement, ensuring optimal airflow, ventilation, and dust minimization

KU10. the methods for installing air cleaners and anti-condensation heaters for different environmental conditions

KU11. the significance of Climate Control Units in maintaining appropriate environmental conditions for telecom equipment
KU12. the structured cabling techniques for passive infrastructure and telecom equipment
KU13. the safe and efficient power distribution methodologies at telecom sites
KU14. the power supply requirements for gNodeB/eNodeB and best practices for their installation
KU15. the importance and methods of heat dissipation for 5G network equipment using heat sinks
KU16. the secure connection protocols for integrating telecom power supply units and RF equipment
KU17. the cable selection and protection techniques to prevent weather-induced wear and damage
KU18. the installation and power connection procedures for 5G base stations, RF transceivers, and RRUs
KU19. the emergency power supply integration methods for uninterrupted 5G network operation
KU20. appropriate protective measures for passive infrastructure equipment, including shelters and telecom ducts
KU21. the electrical safety protocols and usage of PPE for handling power infrastructure
KU22. troubleshooting techniques for resolving power supply and equipment connection issues
KU23. the importance and process of coordinating with manufacturers for equipment servicing and rectifying manufacturing defects

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. document installation details, test results, and equipment configurations
GS2. interpret manufacturer manuals and guidelines for equipment installation
GS3. communicate effectively with team members and stakeholders regarding installation procedures
GS4. plan and organize tasks to ensure smooth execution of installation activities
GS5. collaborate with co-workers to complete setup tasks efficiently
GS6. apply analytical skills to identify and resolve issues in power infrastructure setup
GS7. make quick and effective decisions in case of equipment failures or safety risks
GS8. adhere to workplace safety protocols and follow industry best practices
GS9. stay updated on emerging trends and advancements in 5G infrastructure technology
GS10. manage time efficiently to complete installation tasks within deadlines

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Install passive infrastructure equipment	**15**	**25**	**-**	**10**
PC1. select appropriate passive infrastructure equipment, such as battery bank, Switched-Mode Power Supply (SMPS) unit, Power Interface Unit (PIU), Diesel Generator (DG) set, and Climate Control Unit, based on site requirements	1.5	2	-	1
PC2. conduct tests on passive infrastructure equipment before installation to verify their functionality	1.5	1.5	-	1
PC3. install the battery bank, battery charger, and battery stand as per manufacturer’s guidelines, ensuring sufficient batteries as per power requirements	1	1.5	-	1
PC4. charge and test the battery bank to confirm its ability to charge and discharge properly	1	1.5	-	1
PC5. install the SMPS unit according to manufacturer instructions and verify it provides regulated output voltage	1	1.5	-	1
PC6. install and test the PIU to ensure protection against voltage fluctuations and surges	1	1.5	-	1
PC7. identify a suitable location for the DG set, ensuring proper ventilation, obstruction-free airflow, and minimal dust accumulation	1	1.5	-	0.5
PC8. ensure adequate space for DG set operation, servicing, and ventilation	1	2	-	0.5
PC9. install the DG set following manufacturer instructions and implement air cleaners for dusty environments and anti-condensation heaters for humid conditions	1	2	-	0.5
PC10. install Climate Control Units to maintain optimal temperature and humidity levels for telecom equipment	1	2	-	0.5

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure correct placement of all passive infrastructure equipment, adhering to manufacturer specifications	1	2	-	0.5
PC12. perform structured cabling for installed equipment, ensuring compatibility and adherence to power distribution guidelines	1	2	-	0.5
PC13. establish a power connection between telecom equipment and the power source to ensure safe and continuous operation	1	2	-	0.5
PC14. conduct functionality tests on installed infrastructure equipment and rectify any identified issues	1	2	-	0.5
Set up passive infrastructure with 5G equipment	**15**	**25**	**-**	**10**
PC15. install the power supply unit for gNodeB/eNodeB, ensuring compatibility with the equipment	1.5	2	-	1
PC16. install a heat sink for gNodeB/eNodeB to facilitate heat dissipation and prevent overheating	1.5	1.5	-	1
PC17. connect the gNodeB/eNodeB with the power supply unit and heat sink using appropriate cabling, ensuring secure connections	1.5	1.5	-	1
PC18. establish a secure connection between power supply units and the main power source	1.5	2	-	1
PC19. set up a reliable power connection between the 5G base station and the power source, adhering to equipment power requirements	1	2	-	1
PC20. use appropriate cables for power supply and provide protection against environmental exposure and wear	1	2	-	1
PC21. connect and power up 5G base stations, 5G Radio Frequency (RF) transceivers, and Remote Radio Units (RRUs) following specified power guidelines	1	2	-	1
PC22. integrate emergency power supply systems with 5G network and transmission equipment	1	2	-	0.5

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. install protective shelters and telecom ducts to safeguard passive infrastructure equipment and cabling	1	2	-	0.5
PC24. establish a site-wide power distribution system for consistent power delivery to all cell tower equipment	1	2	-	0.5
PC25. use suitable Personal Protective Equipment (PPE) while handling electrical installations to ensure safety	1	2	-	0.5
PC26. conduct final testing of all equipment post-installation and troubleshoot any operational issues	1	2	-	0.5
PC27. coordinate with equipment manufacturers to address any defects or malfunctions	1	2	-	0.5
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4203
NOS Name	Install and Set up Passive Infrastructure Equipment with the 5G Equipment
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Network (Passive) Installation
NSQF Level	4
Credits	7
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

TEL/N4204: Maintain Passive Infrastructure Equipment

Description

This unit focuses on maintaining the passive infrastructure equipment at 5G network infrastructure sites to ensure the reliability, efficiency, and longevity of network components.

Scope

The scope covers the following :

- Maintain passive infrastructure equipment
- Document and review the maintenance record

Elements and Performance Criteria

Maintain passive infrastructure equipment

To be competent, the user/individual on the job must be able to:

PC1. inspect passive infrastructure equipment, including battery banks, SMPS, PIU, DG sets, climate control units, and surge protection devices, to ensure optimal functionality

PC2. carry out preventive maintenance of passive infrastructure equipment as per the recommended schedule and industry best practices

PC3. verify the correct positioning, connections, and charge-holding capacity of batteries in the battery bank and replace any faulty units

PC4. inspect, test, and replace damaged or worn-out equipment cables to prevent connectivity failures

PC5. conduct diagnostic tests on batteries to assess performance and execute necessary corrective actions

PC6. ensure the SMPS efficiently controls power at high frequencies and supplies stable power to telecom equipment, performing corrective maintenance when needed

PC7. troubleshoot SMPS-related issues such as overheating, overvoltage, and inconsistent power supply

PC8. perform testing on the PIU to validate its ability to handle power fluctuations and surges efficiently, addressing any malfunctions

PC9. diagnose and resolve issues affecting PIU performance to prevent downtime

PC10. perform routine checks on the DG set for issues such as low oil pressure, high water temperature, overload, start failure, short circuit, phase loss, high voltage, and voltage loss

PC11. execute manufacturer-specified troubleshooting procedures for identified DG set faults

PC12. assess climate control units, including HVAC systems, to ensure optimal temperature and humidity control for telecom equipment protection

PC13. use only manufacturer-approved spare parts and maintenance tools to ensure compliance with safety and performance standards

PC14. replace damaged or irreparable passive infrastructure equipment, ensuring proper integration with existing systems

PC15. use remote monitoring tools where applicable for predictive maintenance of passive infrastructure components

Document and review maintenance records

To be competent, the user/individual on the job must be able to:

PC16. maintain accurate records of all maintenance activities, including inspections, repairs, and replacements

PC17. review and analyze maintenance logs to identify recurring equipment issues and implement preventive strategies

PC18. ensure compliance with regulatory standards and company guidelines through proper documentation and periodic audits

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the significance of regular maintenance and its impact on network uptime and efficiency

KU2. preventive maintenance methodologies for passive infrastructure equipment and their role in minimizing failures

KU3. proper battery bank setup, including positioning, wiring, and performance testing

KU4. the impact of damaged or improperly connected cables on equipment performance and the correct methods for replacement

KU5. best practices for battery testing and troubleshooting techniques for charge retention issues

KU6. SMPS functionality, troubleshooting techniques, and energy efficiency considerations

KU7. PIU testing procedures, including voltage regulation and power fluctuation handling

KU8. common faults in DG sets (e.g., fuel system failures, lubrication issues, and electrical malfunctions) and effective troubleshooting techniques

KU9. HVAC and climate control technologies used for maintaining optimal conditions in telecom facilities

KU10. importance of using manufacturer-recommended spare parts and diagnostic tools

KU11. compliance requirements for equipment maintenance, including safety and environmental regulations

KU12. data-driven decision-making for predictive maintenance based on maintenance records

KU13. integration of IoT and remote monitoring solutions for infrastructure maintenance

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain accurate and structured work-related documentation, including checklists and maintenance logs

GS2. read and interpret equipment manuals, schematics, and industry standards

GS3. listen actively to instructions and collaborate effectively with team members

GS4. communicate clearly with vendors, supervisors, and team members regarding maintenance issues and solutions

GS5. prioritize tasks based on urgency and criticality to ensure minimal network downtime

GS6. coordinate with other departments (e.g., operations, engineering) to align maintenance activities with network needs

GS7. use problem-solving techniques to diagnose and resolve infrastructure failures efficiently

GS8. adapt to technological advancements, such as automated monitoring systems, for improved maintenance efficiency

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain passive infrastructure equipment	**25**	**45**	**-**	**13**
PC1. inspect passive infrastructure equipment, including battery banks, SMPS, PIU, DG sets, climate control units, and surge protection devices, to ensure optimal functionality	1.5	3	-	1
PC2. carry out preventive maintenance of passive infrastructure equipment as per the recommended schedule and industry best practices	2	3	-	1
PC3. verify the correct positioning, connections, and charge-holding capacity of batteries in the battery bank and replace any faulty units	2	3	-	1
PC4. inspect, test, and replace damaged or worn-out equipment cables to prevent connectivity failures	2	3	-	1
PC5. conduct diagnostic tests on batteries to assess performance and execute necessary corrective actions	2	3	-	-1
PC6. ensure the SMPS efficiently controls power at high frequencies and supplies stable power to telecom equipment, performing corrective maintenance when needed	2	3	-	1
PC7. troubleshoot SMPS-related issues such as overheating, overvoltage, and inconsistent power supply	1.5	3	-	1
PC8. perform testing on the PIU to validate its ability to handle power fluctuations and surges efficiently, addressing any malfunctions	1.5	3	-	1
PC9. diagnose and resolve issues affecting PIU performance to prevent downtime	1.5	3	-	1
PC10. perform routine checks on the DG set for issues such as low oil pressure, high water temperature, overload, start failure, short circuit, phase loss, high voltage, and voltage loss	1.5	3	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. execute manufacturer-specified troubleshooting procedures for identified DG set faults	1.5	3	-	1
PC12. assess climate control units, including HVAC systems, to ensure optimal temperature and humidity control for telecom equipment protection	1.5	3	-	1
PC13. use only manufacturer-approved spare parts and maintenance tools to ensure compliance with safety and performance standards	1.5	3	-	1
PC14. replace damaged or irreparable passive infrastructure equipment, ensuring proper integration with existing systems	1.5	3	-	1
PC15. use remote monitoring tools where applicable for predictive maintenance of passive infrastructure components	1.5	3	-	1
Document and review maintenance records	**5**	**5**	**-**	**7**
PC16. maintain accurate records of all maintenance activities, including inspections, repairs, and replacements	2	1	-	2
PC17. review and analyze maintenance logs to identify recurring equipment issues and implement preventive strategies	2	2	-	3
PC18. ensure compliance with regulatory standards and company guidelines through proper documentation and periodic audits	1	2	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4204
NOS Name	Maintain Passive Infrastructure Equipment
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Network (Passive) Installation
NSQF Level	4
Credits	3
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

TEL/N9109: Follow sustainable practices in telecom infrastructure management

Description

This unit focuses on sustainability practices in telecom infrastructure management, covering e-waste management, green practices, hazardous and non-hazardous waste, and energy efficiency.

Scope

The scope covers the following :

- Manage hazardous and e-waste at telecom sites
- Adopt green energy and resource efficiency practices
- Follow waste reduction strategies
- Ensure compliance with environmental regulations

Elements and Performance Criteria

Manage hazardous and e-waste at telecom sites

To be competent, the user/individual on the job must be able to:

PC1. identify, segregate, and categorize e-waste (batteries, cables, outdated equipment) and hazardous waste (lead-acid batteries, diesel residues, etc.)

PC2. dispose of or recycle waste following the applicable e-waste management guidelines

PC3. follow safe handling procedures for hazardous materials, including protective gear usage

PC4. maintain logs and records of disposed, recycled, or repurposed telecom waste

Adopt green energy and resource efficiency practices

To be competent, the user/individual on the job must be able to:

PC5. optimize power usage through energy-efficient telecom equipment (e.g., LED lighting, smart cooling systems

PC6. assist in adopting solar-powered telecom towers and integration of hybrid energy systems (e.g. solar power and battery backup)

PC7. monitor and minimize fuel consumption in Diesel Generators (DG) sets through load balancing and regular maintenance

Follow waste reduction strategies

To be competent, the user/individual on the job must be able to:

PC8. reduce packaging waste by promoting the reuse of telecom materials and accessories

PC9. follow rainwater harvesting and wastewater reuse practices in telecom cooling systems

PC10. promote structured cabling to reduce cable waste and optimize resource utilization

Ensure compliance with environmental regulations

To be competent, the user/individual on the job must be able to:

PC11. follow Central Pollution Control Board (CPCB) guidelines for telecom site waste disposal

PC12. assist in conducting periodic environmental audits to ensure sustainability compliance

PC13. guide co-workers on eco-friendly practices and waste management policies at telecom sites

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. e-waste management rules (2022) applicable the telecom sector
KU2. Central Pollution Control Board (CPCB) hazardous waste disposal regulations
KU3. safety standards for battery handling and disposal (lead-acid, lithium-ion
KU4. the importance of structured cabling and cable reuse to minimize telecom waste
KU5. the techniques for energy optimization in telecom infrastructure (e.g., smart cooling, LED lighting, hybrid power systems
KU6. the role of solar energy and other renewable sources in reducing telecom carbon footprint
KU7. green telecom practices like fuel efficiency in DG sets and power-saving measures
KU8. best practices in telecom tower site waste management
KU9. the principles of water conservation and sustainable telecom site design
KU10. the importance of training telecom employees on environmental awareness and compliance

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. explain sustainability practices to team members
GS2. identify and address issues in waste management and energy efficiency
GS3. maintain logs on waste disposal, recycling, and energy savings
GS4. analyze power consumption and propose efficiency improvements
GS5. select between disposal, recycling, or repurposing of telecom waste
GS6. adhere to the applicable environmental laws and guidelines
GS7. work with vendors and waste management partners for sustainable practices
GS8. adapt to new green technologies and sustainability policies
GS9. ensure prompt and efficient waste disposal and as per applicable schedules

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Manage hazardous and e-waste at telecom sites	**8**	**15**	**-**	**5**
PC1. identify, segregate, and categorize e-waste (batteries, cables, outdated equipment) and hazardous waste (lead-acid batteries, diesel residues, etc.)	2	4	-	2
PC2. dispose of or recycle waste following the applicable e-waste management guidelines	2	4	-	1
PC3. follow safe handling procedures for hazardous materials, including protective gear usage	2	4	-	1
PC4. maintain logs and records of disposed, recycled, or repurposed telecom waste	2	3	-	1
Adopt green energy and resource efficiency practices	**6**	**15**	**-**	**5**
PC5. optimize power usage through energy-efficient telecom equipment (e.g., LED lighting, smart cooling systems	2	5	-	2
PC6. assist in adopting solar-powered telecom towers and integration of hybrid energy systems (e.g. solar power and battery backup)	2	5	-	1
PC7. monitor and minimize fuel consumption in Diesel Generators (DG) sets through load balancing and regular maintenance	2	5	-	2
Follow waste reduction strategies	**8**	**10**	**-**	**5**
PC8. reduce packaging waste by promoting the reuse of telecom materials and accessories	3	4	-	2
PC9. follow rainwater harvesting and wastewater reuse practices in telecom cooling systems	3	3	-	2
PC10. promote structured cabling to reduce cable waste and optimize resource utilization	2	3	-	1
Ensure compliance with environmental regulations	**8**	**10**	**-**	**5**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow Central Pollution Control Board (CPCB) guidelines for telecom site waste disposal	3	4	-	2
PC12. assist in conducting periodic environmental audits to ensure sustainability compliance	3	3	-	2
PC13. guide co-workers on eco-friendly practices and waste management policies at telecom sites	2	3	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9109
NOS Name	Follow sustainable practices in telecom infrastructure management
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

KU10. how to compute income and expenses
KU11. importance of maintaining safety and security in financial transactions
KU12. different legal rights and laws
KU13. how to operate digital devices and applications safely and securely
KU14. ways to identify business opportunities
KU15. types of customers and their needs
KU16. how to apply for a job and prepare for an interview
KU17. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively using appropriate language
GS2. behave politely and appropriately with all
GS3. perform basic calculations
GS4. solve problems effectively
GS5. be careful and attentive at work
GS6. use time effectively
GS7. maintain hygiene and sanitisation to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
Constitutional values – Citizenship	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
Basic English Skills	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
Communication Skills	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
Diversity & Inclusion	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
Financial and Legal Literacy	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
Essential Digital Skills	**4**	**6**	**-**	**-**
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
Entrepreneurship	**3**	**5**	**-**	**-**
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
Customer Service	**2**	**2**	**-**	**-**
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**1**	**3**	**-**	**-**
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N4202.Prepare for the Installation of 5G Network Infrastructure	30	50	0	20	100	25
TEL/N4203.Install and Set up Passive Infrastructure Equipment with the 5G Equipment	30	50	0	20	100	25
TEL/N4204.Maintain Passive Infrastructure Equipment	30	50	0	20	100	25
TEL/N9109.Follow sustainable practices in telecom infrastructure management	30	50	0	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	**140**	**230**	**-**	**80**	**450**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
TSSC	Telecom Sector Skill Council
TRAI	Telecom Regulatory Authority of India
SMPS	Switched-Mode Power Supply Unit
PIU	Power Interface Unit
DG	Diesel Generator
RF	Radio Frequency
RRU	Remote Radio Unit
PPE	Personal Protective Equipment
HVAC	Heating, Ventilation, and Air Conditioning
IoT	Internet of Things
LED	Light-Emitting Diode
CPCB	Central Pollution Control Board

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.